

Kunyi (Mark) Ma

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Personal Webpage: markmamathematics.com

Education

Columbia University, New York, NY

Ph.D. Candidate in Mathematics

2024–2029 (to be expected)

M.A. in Mathematics, 4.15/4.

2024–2026

Advisors: Daniela De Silva, Ovidiu Savin

Research interest: PDE Analysis, Free Boundaries

New York University Shanghai, Shanghai, China

B.S. in Honors Mathematics and Data Science, magna cum laude, 3.897/4

2020–2024

Existence of Stationary Measure for Incompressible 2D Stochastic Navier-Stokes Equations(with Vahagn Nersesyan)

On modifying Loss Function for PINO to approach Mathematics Nature in Fluid Mechanic(with Zhuang Su)

Nanyang Model High School, Shanghai, China

2017–2020

Research

Preprints

1. Kunyi(Mark) Ma. *A Short Proof of Optimal Regularity for minimizers of the Alt-Phillips Problem.* arXiv:2606.06916 [math.AP], Submitted, 2026.
2. Nicholas Gismondi, Kunyi(Mark) Ma, Mandon Pathak, Alexandru F. Radu. *Non-unique solutions to the periodic $gKdV$ equation.* arXiv:2606.06916 [math.AP], Submitted, 2026.

Talks

Conference and Workshops

Black Swan Academy of Mathematics: Conference on Geometric Analysis and PDEs, University of Western Australia (60 min invited talk). Aug.20th 2026.

The 15th AIMS Conference, Special Session 140, Athens (30 min invited talk). July.10th 2026.

AMS New England Graduate Student Conference 2026, Brown University (10 min talk). Apr.12th 2026.

Workshop on Harmonic Analysis and Partial Differential Equations, University of Pittsburg (poster presentation). Nov.15th 2025.

Seminars

Seminar on Hyperbolic Conservation Laws, Shanghai Normal University. July 29th 2026.

Graduate Student Analysis Seminar, Purdue University. Apr.28th 2026.

Student Analysis Seminar, Courant Institute NYU. Apr.20th 2026.

Online PDE Coffee Chat (Introduction to Elliptic PDEs I & II). Mar.14th/21st 2026.

Columbia

ELLIPTIC PDE GROUP MEETING

Optimal Regularity for Alt-Phillips N Membrane Problem Apr.1st 2026

Caffarelli-Cabr : Jensen's Approximation and Uniqueness of viscosity solution to Dirichlet BVP Feb.11th 2026

Alt-Caffarelli: Viscosity Solution, Non-degeneracy, Lebesgue Density Estimate Nov.7th 2025

Alt-Caffarelli: Euler-Lagrange Equations, Optimal Regularity Nov.6th 2025

The Obstacle Problem: Singular Set, Monneau Monotonicity, ACF Oct.14th 2025

The Obstacle Problem: Classification of Blowups Sep.30th 2025

The Obstacle Problem: Blowups, 2-homogeneity, Convexity Sep.23rd 2025

The Obstacle Problem: Minimizers, EL equations, Continuity, Optimal Regularity, Nondegeneracy Sep.9th 2025

STUDENT ANALYSIS SEMINAR: BOUNDED VARIATION FUNCTIONS & MINIMAL SURFACE PDE, Fall 2024

De Giorgi's Lemma for C^1 Caccioppoli Sets Nov.6th 2024

De Giorgi's Lemma for Harmonic and C^1 function Oct.16th 2024

Regularity of the reduced boundary Sep.25th 2024

Basic properties of functions of bounded variation Sep.11th 2024

INFORMAL READING SEMINAR ON GR, Fall 2024	
Local Theory for Quasi-linear Wave Equations	Nov.18th 2024
Existence of Solutions to non-constant coefficient linear wave IVP	Oct.7th 2024
Representation formula using Homogeneous Distribution	Sep.16th 2024

Teachings

Teaching Assistant

Applied Functional Analysis APMA4150, Columbia University	Spring 2026
Ordinary Differential Equations UN2030, Columbia University	Spring 2026
Analysis & Probability I GR6151 (Graduate Level), Columbia University	Fall 2025
Linear Algebra S2010, Columbia University	Summer 2025 B
Analysis and Optimization S2500 (Grader), Columbia University	Summer 2025 B
Introduction to Modern Analysis I S4061, Columbia University	Summer 2025 A

Directed Reading Program

Introduction to Elliptic PDEs and Geometric Measure Theory (with Liam Dennehy, Marlet Ma, Barnett Pesin) Spring 2026

Student Tutor

College Algebra and Calculus I–III, Columbia University	Fall 2024
ARC Learning Assistant for Pre-Calculus, NYU Shanghai	Fall 2021

Honors and Awards

NYU Shanghai

Magna cum laude; Phi Beta Kappa; NYU Shanghai Excellence Award; NYU Honors Scholar: Founders Day Award; 2023 NYU Shanghai Dean's Undergraduate Research Funding; 2022 NYU Shanghai Hack the Pearl, Champion of Data Modelling Track; 2020–2022 NYU Shanghai Recognition Award; 2020–2023 NYU Shanghai Dean's List

Attendance (No Talk)

2nd OIST-Oxford-SLMATH Summer Graduate School on Analysis and Partial Differential Equations	June-July 2026
SIAM Conference on Analysis of Partial Differential Equations (PD25)	Nov. 2025
The Third Joint SIAM/CAIMS Annual Meetings (AN25), Montréal, Québec, Canada	July 2025
PDE Summer School 2025 Theory and Applications of Elliptic PDE, UC Irvine	June 2025
Rocky Mountain Mathematics Consortium Summer School 2025, ASU, University of Wyoming	June 2025
Duke Mathematics Journal 2025 Conference, Duke University	Apr. 2025
Winter School: Boundary and Singularity in Fluid Mechanics, Simons Center	Jan. 2025
Differential Geometry Summer School 2024, PKU BICMR	July-Aug. 2024
Workshop on Minimal Surfaces and Mean Curvature Flow 2024, PKU BICMR	July 2024
International Congress of Chinese Mathematicians 2022, THU YMSC	July-Aug. 2022

Extracurricular

Orientation Ambassador, NYU Shanghai	May–Sep. 2021
President, Literature@NYUSH Student Club, NYU Shanghai	Jun. 2021–May 2022
Tenor, NYU Shanghai Elite A Cappella Chamber Singers	Feb. 2021–Dec. 2022
Quantitative Researcher, Ipsos China, Shanghai	Jun.–Aug. 2021